JAVA COLLECTIONS CASE STUDY 1

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Student Marks Management - ArrayList

Topic: List Interface & ArrayList Class
Environment: Java SE 17+ / 24

1. Problem Statement & Scenario Description

Scenario: A teacher wants to maintain the marks of students in a class. The system should allow adding marks, calculating the average, finding the highest mark, and displaying all marks.

#	Required Operation / Method	Implementation Details & Collections Logic
1	**addMarks(List marks, int mark)**	Appends the given mark into the dynamic ArrayList using marks.add(mark).
2	**calculateAverage(List marks)**	Sums all elements using an enhanced for-loop and divides by marks.size(). Returns 0.0 if empty.
3	**findHighest(List marks)**	Iterates through the list tracking the maximum integer value. Returns highest mark found.
4	**displayMarks(List marks)**	Iterates and prints marks separated by single spaces on one line with trailing newline.

2. Collections Framework Concepts & Technical Analysis

- **ArrayList Implementation:** Implements the List interface backed by a resizable dynamic array, providing O(1) amortized insertion.
- **Type Safety with Generics:** Uses List to guarantee homogeneous integer elements and eliminate runtime ClassCastException.
- **Dynamic Aggregation:** Calculates statistical aggregate values (mean average and peak maximum) via sequential list traversal.

3. Complete Java Solution Code (Human Written - No Comments)

Clean, idiomatic Java implementation completed according to instructions with zero comments.

Source File: src/collections_case_studies/StudentMarks.java

```
package collections_case_studies;

import java.util.ArrayList;
import java.util.List;

public class StudentMarks {

    public static void addMarks(List<Integer> marks, int mark) {
        marks.add(mark);
    }

    public static double calculateAverage(List<Integer> marks) {
        if (marks.isEmpty()) {
            return 0.0;
        }
        double sum = 0.0;
        for (int mark : marks) {
            sum += mark;
        }
        return sum / marks.size();
    }

    public static int findHighest(List<Integer> marks) {
        if (marks.isEmpty()) {
            return 0;
        }
        int highest = marks.get(0);
        for (int mark : marks) {
            if (mark > highest) {
                highest = mark;
            }
        }
        return highest;
    }

    public static void displayMarks(List<Integer> marks) {
        for (int i = 0; i < marks.size(); i++) {
            System.out.print(marks.get(i));
            if (i < marks.size() - 1) {
                System.out.print(" ");
            }
        }
        System.out.println();
    }

    public static void main(String[] args) {
        List<Integer> marks = new ArrayList<>();

        addMarks(marks, 78);
        addMarks(marks, 85);
        addMarks(marks, 92);
        addMarks(marks, 67);
        addMarks(marks, 88);

        displayMarks(marks);

        System.out.println("Average: " + calculateAverage(marks));
        System.out.println("Highest: " + findHighest(marks));
    }
}
```

4. Program Execution & Verification Output

Console output produced by running the completed driver code matching expected laboratory output:

Console Standard Output (stdout): java collections_case_studies.StudentMarks

```
78 85 92 67 88
Average: 82.0
Highest: 92
```